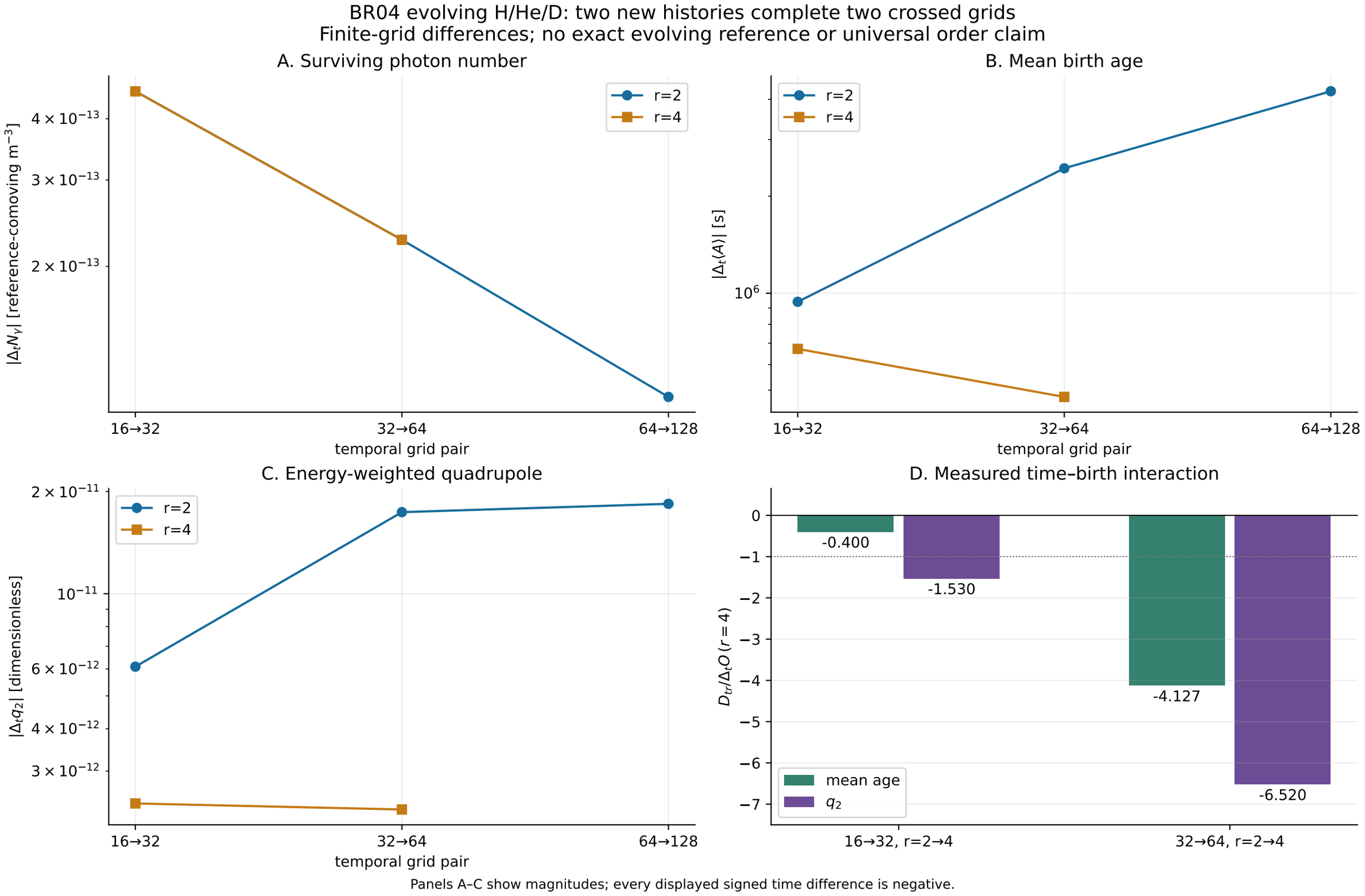
BR04 evolving H/He/D: two new histories complete two crossed grids
Finite-grid differences; no exact evolving reference or universal order claim
A. Surviving photon number
$|\Delta_t N_\gamma|$ [reference-comoving m$^{-3}$]
$4 \times 10^{-13}$
$3 \times 10^{-13}$
$2 \times 10^{-13}$
r=2
r=4
16→32
32→64
64→128
temporal grid pair
B. Mean birth age
$|\Delta_t \langle A \rangle|$ [s]
$10^6$
r=2
r=4
16→32
32→64
64→128
temporal grid pair
C. Energy-weighted quadrupole
$|\Delta_t q_2|$ [dimensionless]
$2 \times 10^{-11}$
$10^{-11}$
$6 \times 10^{-12}$
$4 \times 10^{-12}$
$3 \times 10^{-12}$
r=2
r=4
16→32
32→64
64→128
temporal grid pair
D. Measured time–birth interaction
$D_{tr}/\Delta_t O\,(r=4)$
0
−1
−2
−3
−4
−5
−6
−7
-0.400
-1.530
-4.127
-6.520
mean age
$q_2$
16→32, r=2→4
32→64, r=2→4
Panels A–C show magnitudes; every displayed signed time difference is negative.